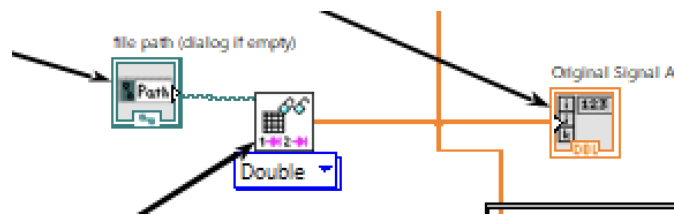


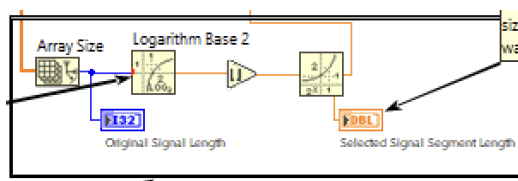
Explanation of Project:

Our program allows the user to enter a file path to an ECG data file, which is then loaded



into a time-domain array. Then the ECG signal is read and determines the total length, which is extracted from the

corresponding segment from the start of the signal. This fixed-length segment ensures that all



signals are processed with a consistent input size prior

to wavelet analysis. After that, the raw ECG signal is

displayed using a waveform graph. The selected

segment is transformed using a Daubechies 4 wavelet

transform, which decomposes the signal into its

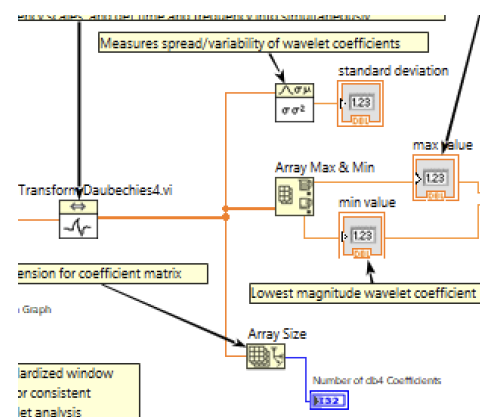
components and outputs a coefficient matrix. From the

wavelet coefficient matrix, statistical features including

the maximum, minimum, range (max – min), and

standard deviation are computed. This feature-reduction

step decreases the dimensionality of the wavelet output



while preserving distinguishing characteristics of each ECG signal.

The extracted feature values, particularly the coefficient range, are

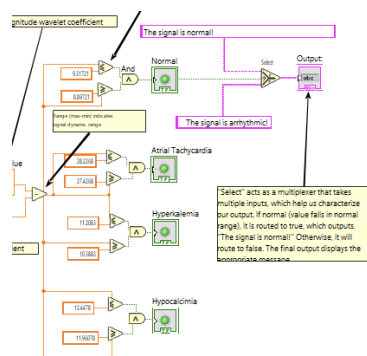
then compared with predefined thresholds to classify each potential

heart condition. Classification results are represented as true/false

boolean indicators for normal rhythm, atrial tachycardia,

hyperkalemia, and hypocalcemia, which are then used to generate

a corresponding text output indicating the classified signal type (normal or arrhythmic).



Background:

Heart disease is a leading cause of death worldwide, resulting in approximately 17.1 million lives globally per year (Sahoo). Among the many cardiovascular diseases, cardiac arrhythmias are dangerous because of their impact on the heart's biological ability to regulate an effective and constant rhythm (Dhande). An arrhythmia can be defined as an irregular heartbeat where the heart rate may become excessively slow, fast, or erratic. After enough time, the irregularity in the heartbeat can result in not enough blood being pumped throughout the body. If left untreated, it can cause a variety of complications to occur, such as heart failure, stroke, or cardiac arrest (Nagpal). Long-term irregularities in heart rhythm can lead to a lack of blood and oxygen in vital organs, thereby limiting cardiac output.

An electrocardiography (ECG) is a non-invasive test used to record the heart's electrical activity in order to determine the heart's rhythm, rate, and overall function (Rafie). The physician will assess by utilizing electrodes in locations such as the chest, arms, and legs to measure and graph activity. Effective analysis of ECG signals provides life-saving information on how to effectively treat patients through understanding the patterns that occur in the heart's activity. For arrhythmias, ECGs are the most commonly used detection tools through the identification of atrial fibrillation on the ECG (Mahmoodabadi). The ECG can detect the rhythms and localize the origin of the rhythm in many cases, as this displays the importance of understanding the computational analysis of ECG signals for early detection and classification of arrhythmias.

LabVIEW is utilized in biomedical engineering as it is able to process and analyze signals due to its programming environment and the toolsets that allow for customized dataflow structures. In particular, the Wavelet Analysis Tool provides a set of toolkits to assist with processing signals in the LabVIEW environment.

The wavelet transform is a tool in LabVIEW that can be used to break up the ECG signal into short-term details and long-term trends. The wavelet transform can capture the overall trend of the rhythm of a heartbeat, as well as the short, sharp peaks that can appear on an ECG. From this, the wavelet transform extracts information that distinguishes a normal heart rhythm from an arrhythmia. Therefore, wavelet-based analysis provides features that can be used to distinguish normal heart rhythms from certain arrhythmic conditions. From the coefficients generated by the wavelet, you can determine the mean of the coefficients, peaks related to ECG components, and the variance (Sahoo). The variance in these coefficients provides quantitative features that can be used to distinguish between normal and abnormal heart rhythms. The wavelet transform generates both time and frequency information, unlike the traditional Fourier transform, which provides frequency averaged over time. Because ECG signals are non-stationary and exhibit time-varying characteristics, wavelet-based analysis is particularly well-suited for feature extraction in this application.

Our strategy for classifying the ECG signals was to extract statistical features from the Daubechies 4 wavelet coefficient matrix, including the maximum, minimum, range, and standard deviation. These reduced features were then used to distinguish normal from abnormal heart rhythms using a threshold-based classification.

Sources:

Source 1:

S. Sahoo, B. Kanungo, S. Behera, and S. Sabut, "Multiresolution wavelet transform based feature extraction and ECG classification to detect cardiac abnormalities," *Measurement*, vol. 108, pp. 55–66, Oct. 2017, doi: <https://doi.org/10.1016/j.measurement.2017.05.022>. Available: <https://www.sciencedirect.com/science/article/pii/S026322411730297X>. [Accessed: Jan. 28, 2026]

Source 2:

Nagpal, Anmol K et al. "Cardiac Arrhythmias and Their Management: An In-Depth Review of Current Practices and Emerging Therapies." *Cureus* vol. 16,8 e66549. 9 Aug. 2024, doi:10.7759/cureus.66549

Source 3:

S. Dhande, "LABVIEW BASED ECG SIGNAL ACQUISITION AND ANALYSIS," *International Journal of Engineering Applied Sciences and Technology*, vol. 5, no. 9, Jan. 2021, doi: <https://doi.org/10.33564/ijeast.2021.v05i09.046>. Available: <https://www.ijeast.com/papers/285-290%2CTesma509%2CIIJEAST.pdf>. [Accessed: Jan. 28, 2026]

Source 4:

Rafie, Nikita, et al. "ECG Interpretation: Clinical Relevance, Challenges, and Advances." *Hearts*, vol. 2, no. 4, 2 Nov. 2021, pp. 505–513, www.mdpi.com/2673-3846/2/4/39, <https://doi.org/10.3390/hearts2040039>.

Source 5:

S. Z. Mahmoodabadi, A. Ahmadian, M. D. Abolhasani, M. Eslami, and J. H. Bidgoli, "ECG Feature Extraction Based on Multiresolution Wavelet Transform," *PubMed*, Jan. 2005, doi: <https://doi.org/10.1109/iembs.2005.1615314>. Available: <https://ieeexplore.ieee.org/document/1615314>. [Accessed: Jan. 28, 2026]

Source 6:

Latif, Israa. (2025). "Comparative Study of Discrete Wavelet Transforms in Image Processing using LabVIEW 2023". *Mustansiriyah Journal of Pure and Applied Sciences*. 3. 151-172. 10.47831/mjpas.v3i3.266.

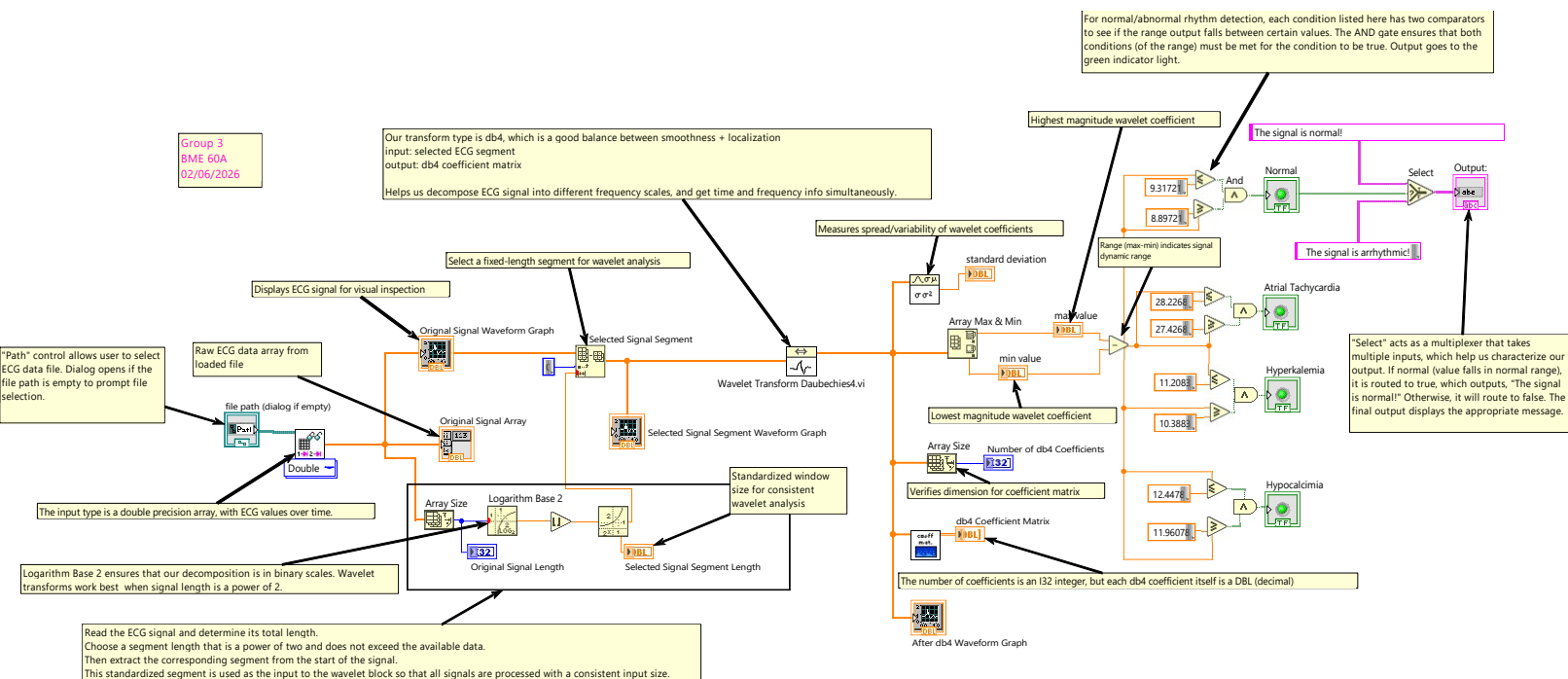


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file path (dialog if empty)

Original Signal Waveform Graph

Plot 0

Amplitude

Time

Original Signal Array

Original Signal Length

Selected Signal Segment Waveform Graph

Plot 0

Amplitude

Time

Selected Signal Segment Length

After db4 Waveform Graph

Plot 0

Amplitude

Time

Number of db4 Coefficients

db4 Coefficient Matrix

standard deviation

max value

min value

Normal

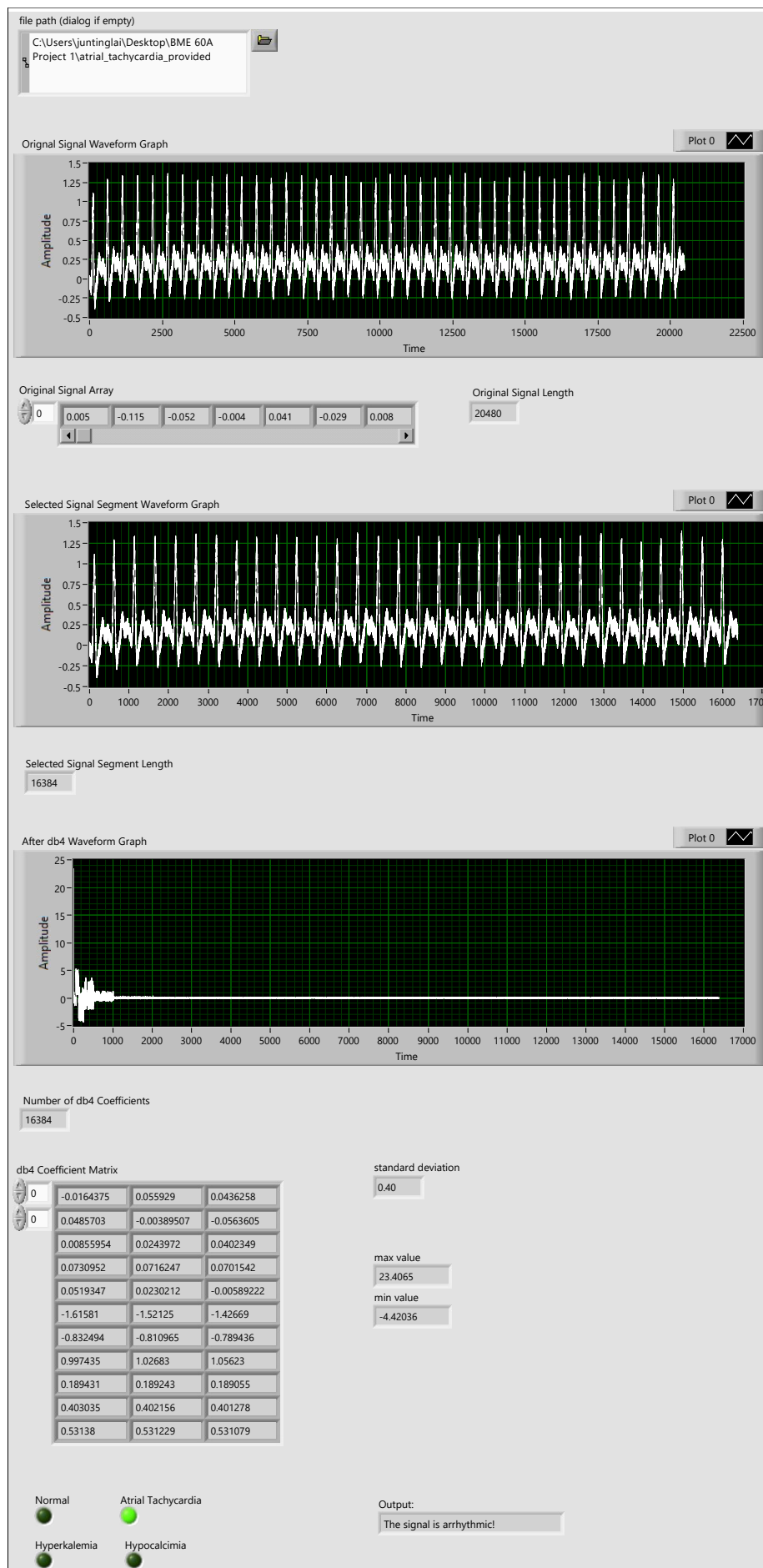
Atrial Tachycardia

Hyperkalemia

Hypocalcemia

Output:

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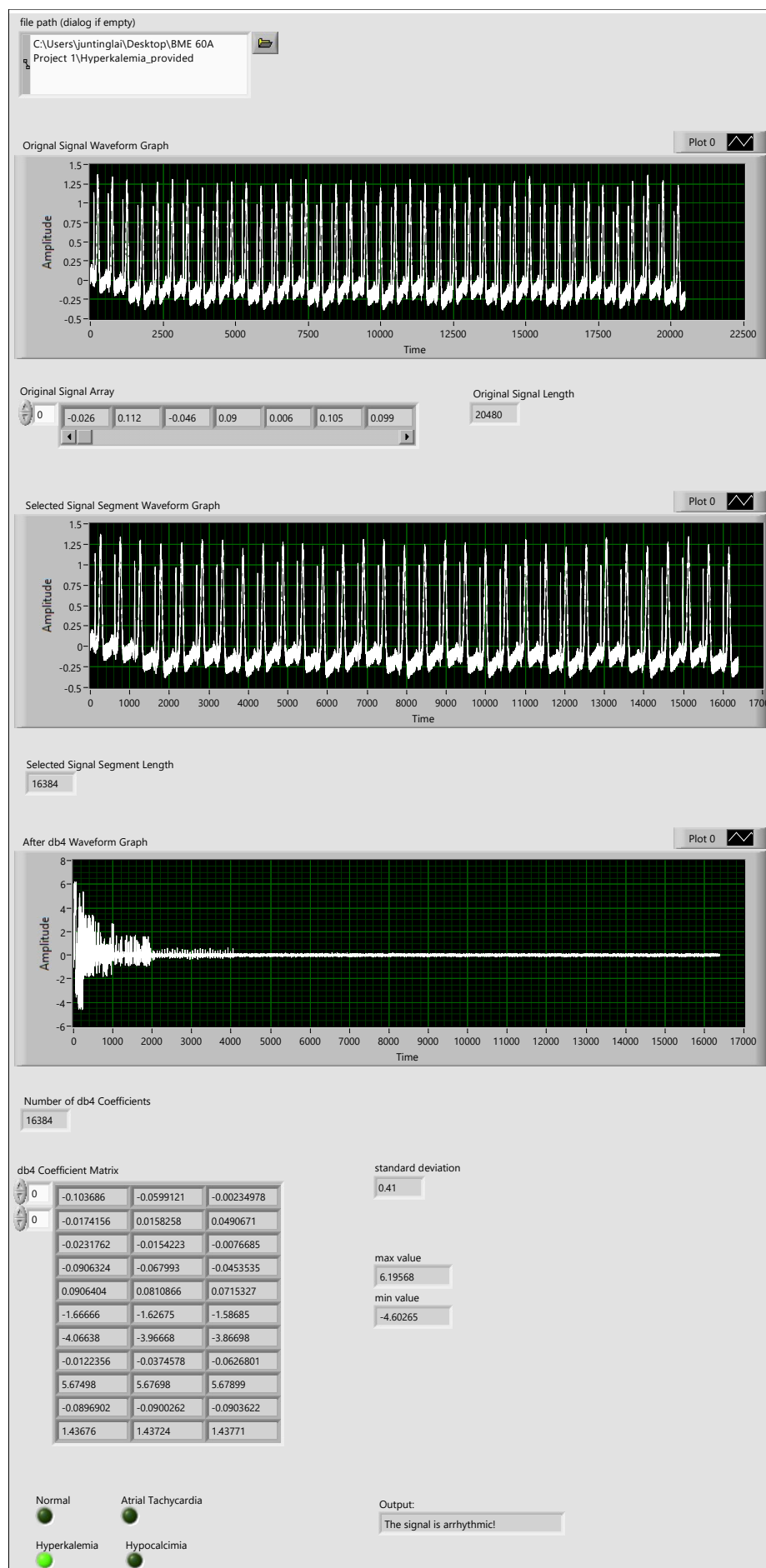


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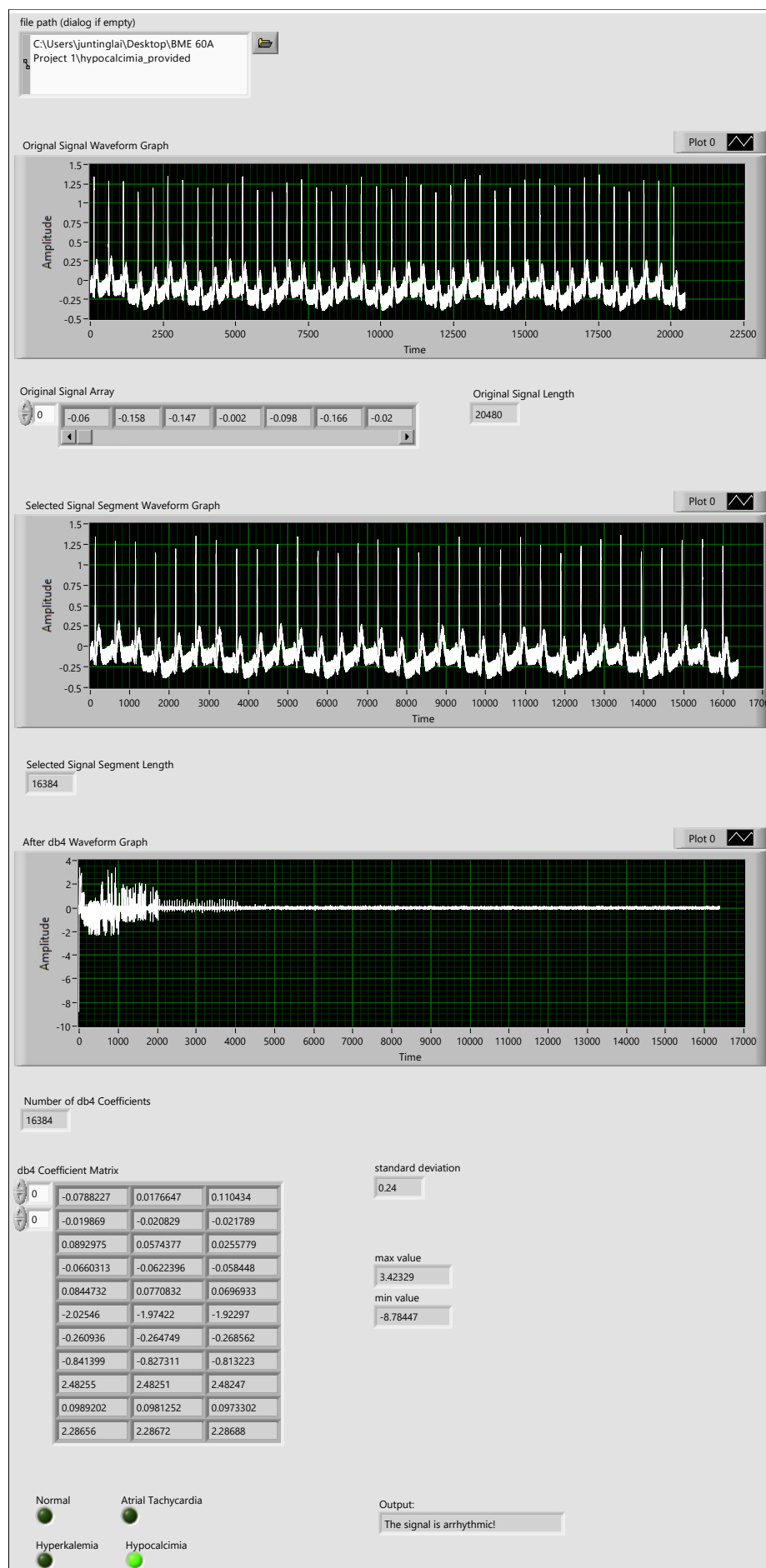


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